

Assignment 03: Implementing Linear Regression and Gradient Descent Optimization

CSci 574: Machine Learning

Description

Introduction

Welcome to our assignment on implementing linear regression and gradient descent from scratch. In this assignment you will build, in Python, functions you need to perform a linear regression optimization by hand. This includes implementing the functions to calculate predictions given a set of weights for a model (they model hypothesis), the mean squared error (MSE) loss function, which we will use to determine the “goodness” of solutions, and the gradients of model weights with respect to this cost, so that you can perform simple gradient descent optimization.

It is useful to build basic linear and logistic regression implementations by hand, at least once, to help you build up your intuition of how parametric models like linear and logistic regression actually work, and how optimization using a model’s gradients is performed.

Instructions:

- You need to use the class development environment and make sure that you are pushing your assignments to your GitHub classroom and they are successfully passing the autograder.
- Avoid using for-loops and while loops, unless you are explicitly told to do so in the assignment task functions for this assignment.
- The `### TESTED FUNCTION [function name]` cells call unit tests on the functions you are asked to write for these assignments. In this assignment, you need to uncomment the code in these cells and add in the needed function as described for each task.
- All functions you need to write should be placed into the `src/assignment_tasks.py` file. Functions that are tested and graded are imported from there into this notebook.
 - You have not been given function documentation and stub functions for this and future assignment. You need to add in the function declarations and uncomment the code to import and test your functions.
 - You are required to give NumPy style formatted Pydoc documentation for your functions.
- After coding your function, run the `### TESTED FUNCTION` cell to determine if it is passing the assignment unit tests and that your result is correct.

Objectives

After this assignment you will:

- Know how to build the general architecture of a learning algorithm for a parametric model, including:
 - Initializing parameters
 - Calculating the cost function and its gradient
 - Using an optimization algorithm (gradient descent)
- Gather all of the above functions into a main model function, in the right order, to perform linear regression on a set of data.

Let’s get started!

Overview and Setup

Assignment Tasks

Assignment Submission

Additional Information